Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

Sea Ice Thematic Assembly Centre

SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_007

Issue: 1.1

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Approval date by the CMEMS product quality coordination team: 05/11/2025

CHANGE RECORD

When the quality of the products changes, the QuID is updated, and a row is added to this table.

The third column specifies which sections or sub-sections have been updated. The fourth column should mention the reason for updating the document. Column 5 should include the names of the author(s) doing the changes. Including the name in column 6 shows that the document has been validated by the TAC/MFC PQ leader or deputy (or TAC/MFC PQ leader or the deputy). This validation means the document was checked for contents, formats and consistency with other documents.

Issue	Date	§	Description of Change	Author	Validated By
1.0	June 2023		Creation of the document, split from the QUID for all the SITAC products. The previous document including this product was CMEMS-SI-QUID-011-001to007-009to015-019-V2.17.	J. Buus-Hinkler	C. Wettre
1.1	May 2025		Expansion of geographical extent to pan-Arctic. Addition of three new datasets based on RCM. Figure 2 (on iceberg detection near the ice-edge) added. Document adapted to the most recent template.	J. Buus-Hinkler	C. Wettre

ACRONYM TABLE

Acronym	Description
AIS	Automatic Identification System
CFAR	Constant False Alarm Rate
CNN	Convolutional Neural Networks
DIDA	DMI Iceberg Detection Algorithm
DMI	Danish Meteorological Institute
DMI-ASIP	DMI Automated Sea Ice Products
EW	Extra Wide Swath (Sentinel-1 acquisition mode)
HH	Horizontal polarization transmit, Horizontal polarization receive (when considering radar signals)
HV	Horizontal polarization transmit, Vertical polarization receive (when considering radar signals)
IW	Interferometric Wide Swath (Sentinel-1 acquisition mode)
NetCDF	Network Common Data Form
NRT	Near Real Time
PC	Production Centre
PU	Production Unit
RCM	Radarsat Constellation Mission
RCMLN	RCM Low Noise (RCM acquisition mode)
SAR	Synthetic Aperture Radar
SI TAC	(Copernicus Marine Service) Sea Ice Thematic Assembly Centre
WMO	World Meteorolocical Organization
NRT	Near Real Time

KEY DEFINITIONS

Term	Definition
AIS	Automatic Identification System, a maritime safety system that ships use to automatically inform about their location, course, and other data. Other ships in range (typically 40-60 nautical miles, depending on antenna height and weather) can receive the signals. Coastal base stations pick up AIS signals near ports and coastlines. Satellites with AIS receivers can capture signals over the open ocean, providing near-global coverage.

CFAR	Constant False Alarm Rate (CFAR) detection is a method used in Synthetic Aperture Radar (SAR) imagery to automatically detect targets like icebergs, ships, or other objects against cluttered backgrounds such as e.g. the sea surface. Instead of using a fixed threshold, CFAR adapts the threshold locally.
CNN	A Convolutional Neural Network is a type of machine learning model - specifically a deep learning neural network - designed to analyze grid-like data such as images.

RELEVANT DOCUMENT LIST

PUM	https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-SEAICE-PUM-011-007.pdf
SQO	https://catalogue.marine.copernicus.eu/documents/SQO/CMEMS-SEAICE-SQO-011-007.pdf

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I

I EXECUTIVE SUMMARY

I.1 Products covered by this document

This document (SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_007) describes the Iceberg concentration (iceberg count per 10 km x 10 km) and individual icebergs observed with Sentinel-1 and Radarsat Constellation Mission (RCM) Synthetic Aperture Radar (SAR) imagery. The product is within the Sea Ice Thematic Assembly Centre (SI TAC) and covers Greenland waters and adjacent Arctic regions for the period 2017–2024, based on Sentinel-1A and Sentinel-1B SAR data, with Sentinel-1B lost in late December 2021 and Sentinel-1A continuing coverage. Sentinel-1C is now included in the production, and Radarsat Constellation Mission (RCM) data have been integrated to reduce coverage gaps. Since November 2023, point-based products have provided pan-Arctic coverage (i.e., all areas where icebergs may occur north of 40° N), and as of November 2025 this has been extended to also include pan-Arctic gridded products in Network Common Data Form (NetCDF) format.

I.2 Summary of the results

Statistical analyses on all our iceberg observations acquired in 2019 reveal that the two main sources of error are: (1) the presence of ice floes (false iceberg-targets) near the sea-ice edge and (2) ships misjudged by the detection algorithm as iceberg-targets.

The iceberg score—an index reflecting the likelihood that a detected target is a true iceberg—is used to analyze detections in relation to their distance from the sea ice edge. The analysis includes data from the full year 2019 and shows that the fraction of significant true iceberg-targets increases with distance from the sea-ice edge up to about 150 km, beyond which it remains constant. This suggests that uncertain targets are present throughout the area but are much more prevalent near the ice edge due to the presence of ice floes.

Targets with high length-to-width ratios are compared to a marine traffic density map. The comparison reveals that areas with dense ship traffic often coincide with regions where targets with high length-to-width ratios are detected. Although this relationship alone is not sufficient to discriminate between ships and icebergs, it helps explain some differences between the two and is confirmed through manual quality checks.

The **2023** version of the DMI Iceberg Detection Algorithm (DIDA) (launched on 30 November 2023) introduced an automated sea-ice filter based on DMI Automated Sea Ice Products (DMI-ASIP), which uses Convolution Neural Networks (CNN) to estimate sea-ice concentration in SAR imagery. Combined with real-time Automatic Identification System (AIS) ship tracking integrated into the chain, this significantly reduces false detections caused by ships and by sea ice floes near the ice edge. These updates are expected to improve detection accuracy, though the full impact has yet to be quantified. A detailed evaluation will be included in the next QUID update in 2026, based on the expanded dataset made possible by the **2025** pan-Arctic extension in geographical coverage and the inclusion of RCM data.

The **2025** version of DIDA also incorporates imagery from RCM, starting with the Low Noise (RCMLN) acquisition mode. RCMLN was introduced first because benchmarking against Sentinel-1 IW reference data showed it produced the fewest obviously false targets compared to other RCM modes. This high reliability made it suitable for direct integration into the DIDA processing workflow. With further studies on noise correction we expect to include more acquisition modes over the coming years. Furthermore, by May 2025, Sentinel-1C imagery was smoothly incorporated in the processing chain.

I.3 Estimated Accuracy Numbers

Table 1 presents the main validation results for the DIDA iceberg detection product, including comparisons with RCM imagery. The definitions of the metrics, including the iceberg score, are described in Sections III and IV.

Table 1. Summary of validation results for the DIDA iceberg detection, including the 2025 update with RCM data integration.

Metric /Test	Result/Key Finding	Implication	Reference
Iceberg score vs. distance to ice edge.	Mean iceberg score increases with distance from ice edge, saturating at ~150 km. ~75% of increase within first 50 km, ~90% within 100 km.	Detection reliability is lowest near the ice edge due to ice floes; false positives most prevalent in marginal ice zone.	Sec. IV; Fig. 1; **Iceberg score defined in Sec. III.1**
Over-detection under extreme settings (extrem_lev=3 vs. 1)	Some grid cells show >10× (up to 100×) more detections under extreme settings, mainly near ice edge.	Confirms that ice floes drive strong over-detection; i.e. need for robust ice-masking.	Sec. IV; Fig. 2A
Sea-ice filtering (DMI-ASIP vs. traditional charts)	DMI-ASIP product removes many false detections that remain with traditional charts; more temporally and spatially consistent.	Demonstrates that positives near the ice edge. DMI-ASIP filtering significantly reduces false detections.	Sec. IV; Fig. 2B
Ship discrimination (length-to-width ratio, >2.5 = likely ship)	Many high-ratio detections coincide with dense shipping lanes; manual checks confirm correspondence with ships.	Length-to-width ratio helps flag likely ships, though not reliable as standalone discriminator.	Sec. IV; Fig. 3
AIS traffic density comparison	Spatial overlap between AIS-derived ship traffic density and high ratio detections.	Supports validation of ship vs. iceberg discrimination approach.	Sec. IV; Fig. 3
RCM Low Noise (RCMLN) vs Sentinel-1 IW (2025 update)	~60% of cases match Sentinel-1 IW reference; ~40% fewer detections (expected due to lower resolution); ~1% false alarms.	RCMLN provides good agreement and minimal false positives; can be integrated into DIDA with no parameter adjustments.	Sec. IV; Table 3a
RCM ScanSARMedium-50 vs Sentinel-1 IW (2025 update)	Only ~19% of cases match Sentinel-1 IW reference; >80% show missing detections; 20–22% false alarms.	ScanSARMedium-50 performs poorly under current settings; requires significant algorithm adjustments before use in DIDA.	Sec. IV; Table 3b

II PRODUCTION SYSTEM DESCRIPTION

The product SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_007 provides iceberg number densities on a 10 × 10 km grid as well as individual iceberg positions with information on size and detection significance (iceberg score). Derived from SAR imagery using the DIDA system, the product is produced by the unit in Copenhagen, Denmark, and distributed through the production centre in Oslo, Norway, see Table 2 below.

Table 2 - Product names, description and system version, SI TAC production unit and production centre

Product	Product description	Version	Production Unit (PU)	Production Centre (PC)
SEAICE_ARC_SEAICE_L4_NRT _OBSERVATIONS_011_007	The product consist of Iceberg Number Densities (number of icebergs in 10x10 km gridcells) and Individual Iceberg postions with information on iceberg size an detection significance (iceberg score). All infeered from SAR imagery using the DIDA-system	2.14	SIW-DMI-COPENHAGEN-DK	SIW-METNO-OSLO-NO

Description

The "Icebergs Observed with Sentinel-1 & RCM – Number Density and Individual Icebergs" products (SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_007) are based on target (iceberg) detection in data inferred from Sentinel-1 and RCM Synthetic Aperture Radar (SAR) imagery processed at the Danish Meteorological institute (DMI). Until September 2023 Sentinel-1 data have been downloaded from the European Space Agency (ESA) ftp servers dedicated for Copernicus service providers to the production unit at DMI. From September 2023 download has changed to the Sentinel-1 data access environment[1]

The products only include icebergs detected in open-water areas (i.e., areas with no sea ice coverage) in the Greenland Waters and include nine datasets based on Sentinel-1- and RCM SAR-scenes. Sentinel-1 imagery is acquired in EW and IW modes, respectively, whereas RCM imagery currently only includes RCMLN acquisitions. The nine datasets consist of six sets of gridded "iceberg number density" available in netCDF4 format and three sets of "individual iceberg positions" available in ESRI Shapefile format:

[1] https://dataspace.copernicus.eu

1. single scenes (EW) **[gridded 10x10 km, netCDF format]**
2. 4-day mosaics (EW) **[gridded 10x10 km, netCDF format]**
3. single scenes (RCMLN) **[gridded 10x10 km, netCDF format]** (new Nov-2025)
4. 4-day mosaics (RCMLN) **[gridded 10x10 km, netCDF format]** (new Nov-2025)
5. single scenes (IW) **[gridded 10x10 km, netCDF format]**
6. 7-day mosaics (IW) **[gridded 10x10 km, netCDF format]**

7. 4-day mosaics (EW) **[point positions, Shapefile format]**
8. 4-day mosaics (RCMLN) **[point positions, Shapefile format]** (new Nov-2025)
9. 7-day mosaics (IW) **[point positions, Shapefile format]**

Method

As a first step DIDA utilizes a Constant False Alarm Rate (CFAR) concept for target detection (see Gill, 2001; Buus-Hinkler et al. 2014, and Karvonen et al., 2022). Basically, this means that if the backscatter value of a given pixel is so high that it is extremely unlikely to belong to the statistical distribution of the nearby background-pixels, this pixel will be classified as a “target-pixel”. After running CFAR-detection, then post-processing routines are applied to identify “true” iceberg-targets and eliminate false targets. This includes analysis of target length to width ratios of the pixel clusters making up the individual targets – plus further analysis of detection significance- and occurrence of target-pixels in the two different polarizations.

A main source of error in the iceberg detection arises from the misclassification of sea ice floes near the ice edge as icebergs by the CFAR detection algorithm. This issue is significantly reduced using DMI-ASIP to mask out sea ice-infested areas in the SAR imagery prior to detection. The key advantage of using DMI-ASIP lies in its timeliness and geometric consistency: unlike traditional ice charts (used in earlier versions of DIDA), which may be issued hours after satellite overpass, DMI-ASIP is generated automatically and based on the same SAR imagery used in the iceberg detection. This ensures a perfect spatial and temporal match, improving the accuracy of the sea ice masking and reducing the risk of false iceberg detections near the dynamic and often rapidly changing ice edge.

To help users better assess *iceberg number densities*, the number-density netCDF-files include statistical layers describing the level of extremity for each individual grid-cell. These layers show the degree to which observations deviate from the normal/prevailing conditions (4-year iceberg climatology for the current season) at the observed spot/grid cell. There are two different statistical layers: 1. extremity index, and 2. the percentile (quantile) that corresponds to the current observation. Grid-cells with high extremity indexes should be treated critically.

Generally, ships appear as significant targets in SAR imagery. Particularly in regions with more dense marine traffic the difficulty in distinguishing between ships and icebergs is a problem. Finally, noise in SAR data (e.g., speckle, sea-clutter, high winds, etc.) in some cases leads to the detection of false iceberg-targets.

III VALIDATION FRAMEWORK

Given the variability in the nature of the Sea Ice Thematic Assembly Center (SI TAC) Sea ice products in terms of the way they are produced and limited availability of in situ reference data, the validation procedures are not standardized. Due to this reason, the SI TAC sea ice validation is currently not able to follow the standard conventions laid out in the Copernicus Marine Environment Monitoring Service (CMEMS) Cal/Val guidelines.

The number of ground-based iceberg observations suitable for robust quality assessment is limited. Therefore, the product quality is evaluated using indirect statistical methods. Specifically, the observations and gridded statistics are compared to: (i) iceberg score versus distance to the sea-ice edge and the occurrence of exceptionally high iceberg concentrations—shown to be false detections caused by ice floes near the ice edge (Figures 1–2); and (ii) ship statistics derived from a marine traffic density map (Figure 3).

III.1 Details on iceberg score

The iceberg score ($Score_{iceberg}$) is an index developed to indicate the likelihood that the detected target is a true iceberg-target.

The index is based on the significances (Sig_{HH} and Sig_{HV}) with which the target in question is detected in HH and HV polarization, respectively. The significance is defined as the percentage at which the backscatter-intensity of the target pixel is above a threshold derived using Constant False Alarm (CFAR) technique – other details on CFAR can be found in the Product User Manual (PUM) for SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_007. Significances may vary from a few percent for insignificant targets to several thousand percent for very bright targets.

As a part of the derivation of $Score_{iceberg}$ the normalized difference ratio between the local HH and HV background intensity is calculated. This normalized difference ratio is defined as:

$$(i_{HH} - i_{HV}) / (i_{HH} + i_{HV}) \text{ , where}$$

i_{HH} and i_{HV} are the backscatter intensities of the background pixels around each target in HH and HV polarization, respectively. In areas where this normalized difference is close to- or below zero, HH backscatter intensities are very low (i.e., close to zero) which may lead to high contrasts even between weak targets (i.e., noise or sea-clutter), which again may lead to many false detections.

Naturally, if a target is detected both in the HH-and HV polarization-channels, it is more likely to be a real target than if detected only in one of them.

Thus, the score is calculated as a weighted sum of Sig_{HH} and Sig_{HV}. According to the above description, the weight of Sig_{HH} should be reduced in areas with low values of $(i_{HH} - i_{HV}) / (i_{HH} + i_{HV})$.

The final iceberg score is thus defined as:

$$Score_{iceberg} = SigHH\sqrt{(i_{HH} - i_{HV})/(i_{HH} + i_{HV})} + SigHV$$

Targets with $Score_{iceberg} < 40$ are considered as false targets and are thus excluded.

Note: The *square root* of the normalized difference is empirically introduced to avoid multiplying SigHH with too low numbers when "$(i_{HH} - i_{HV}) / (i_{HH} + i_{HV})$" approaches zero (i.e. when i_{HH} and i_{HV} are close to be equal).

IV

IV VALIDATION RESULTS

As mentioned in section III two main sources of error are: (1) the presence of ice floes (false iceberg-targets) near the sea-ice edge and (2) ships misjudged by the detection algorithm as iceberg-targets. Thus, the 2023+ versions of the iceberg detection scheme includes a new automated sea ice filter based on DMI's Automated Sea Ice Products (ASIP). The ASIP DMI-ASIP and iceberg products are derived from the same satellite radar scenes, ensuring precise alignment in both time and space. Furthermore, new post-processing routines now combine real-time AIS ship tracking data with the incoming Sentinel-1 imagery, which makes us able to significantly reduce "ship-targets". Therefore, statistics based on future products is expected to reveal less prevalence of these errors.

In section IV.1, the prevalence of these errors is assessed using an empirically derived iceberg score (Figure 1), statistical analysis of extremity (Figure 2) and a maritime traffic density map (Figure 3).

IV.1 Results - Main sources of error and iceberg score

The detection algorithm involves steps to derive the iceberg score, which is an index developed to indicate the likelihood that the target in question is a "true target". Further details of the iceberg score index can be found in section III.1.

Figure 1 compares the iceberg score of detected targets to their distance to the sea-ice edge. The analysis includes one year (2019) of observations (more than ten million) and shows iceberg scores (mean, P16, median, P84) of all targets present within distances ranging from 0 to 500 km from the ice edge. For example, the mean iceberg score of all targets observed during 2019 present within a distance of 100 km from the ice edge is approximately 460 (black curve) and the lowest 16 percent of the iceberg scores within this distance are all lower than about 60 (blue curve).

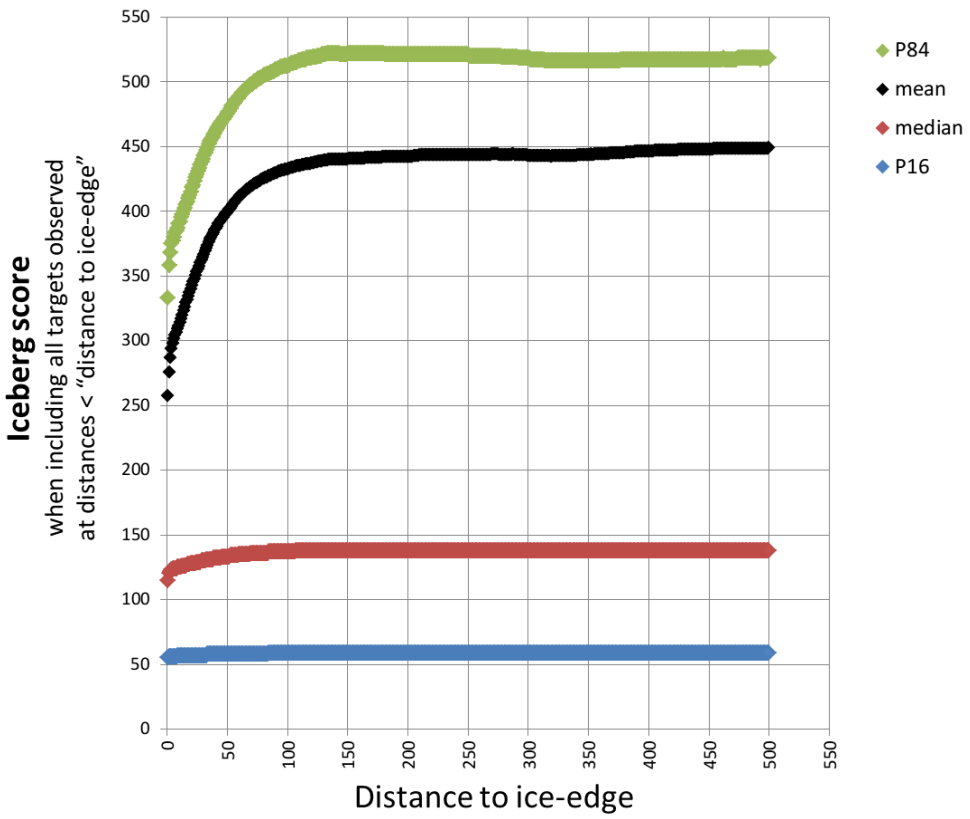

Figure 1. Iceberg score vs. distance to sea-ice-edge. Statistics based on all detections encountered in 2019.

Generally, Figure 1 shows that the fraction of significant-iceberg targets (with high scores) increases with distance from the sea-ice edge up to about 150 km. Beyond 150 km the fraction becomes constant. About 75% of the increase happens within the first 50 km and within 100 km distance 90% of the increase has happened. Thus, uncertain targets are present everywhere, but they are much more prevalent near the ice edge. This confirms the hypothesis that due to the presence of ice floes in the marginal sea-ice zone, iceberg detection is much more uncertain in the vicinity of the sea-ice edge (i.e., the risk of detecting false targets is much larger there). Figure 2 illustrates key challenges in iceberg detection near the sea-ice edge. Figure 2A identifies 10 × 10 km grid cells where the number of detected targets under extreme detection settings (i.e., extrem_lev=3) is more than 10 times — and in some cases up to 100 times — higher than under normal conditions (extrem_lev=1). These areas are primarily located near the ice edge, suggesting that the presence of sea-ice floes may lead to substantial over-detection. Figure 2B compares the effect of using the DMI-ASIP sea-ice product with a traditional ice chart. Due to the time lag between satellite acquisition and the issuance of traditional charts, many false targets near the ice edge are not removed when traditional charts are used. In contrast, the DMI-ASIP product — derived from the same satellite data as the iceberg detections — provides a more temporally and spatially consistent mask. This significantly improves the filtering of sea-ice-contaminated areas and reduces false positives.

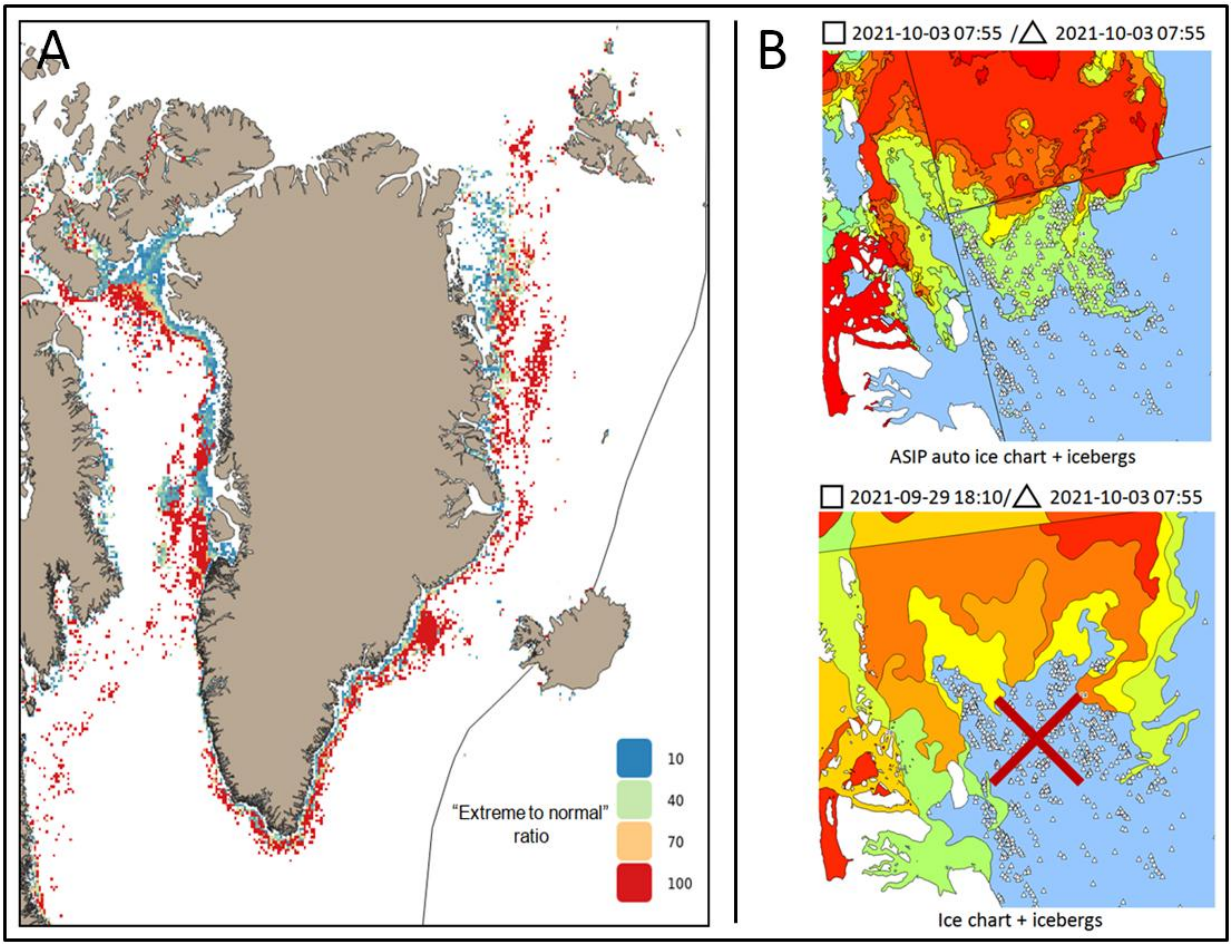

Figure 2. Grid-cells (10 x 10 km) in which the number of detected iceberg-targets is more than 10 times higher under "extreme" conditions (extrem_lev=3) than under "normal" conditions (extrem_lev=1) (A). The figure combines both EW- and IW acquisition modes. DMI-ASIP chart vs. traditional ice chart with iceberg detections (B). Due to the time gap between satellite acquisition and the ice chart many false targets are not removed when applying the traditional chart for sea-ice filtering. Note that in the ice charts sea-ice concentration follows World Meteorological Organization (WMO) standard colors, ranging from open water (blue) through medium concentrations (green–yellow–orange) to very close pack ice (red).

Another main source of falsely detected targets is the presence of ships. These often have high target scores but usually differ in shape (are more elongated than icebergs). Figure 3 compares targets with high "length-to-width-ratios" (Figure 3A) to a marine Traffic Density Map (Figure 3B) covering regions around- and south of Greenland.

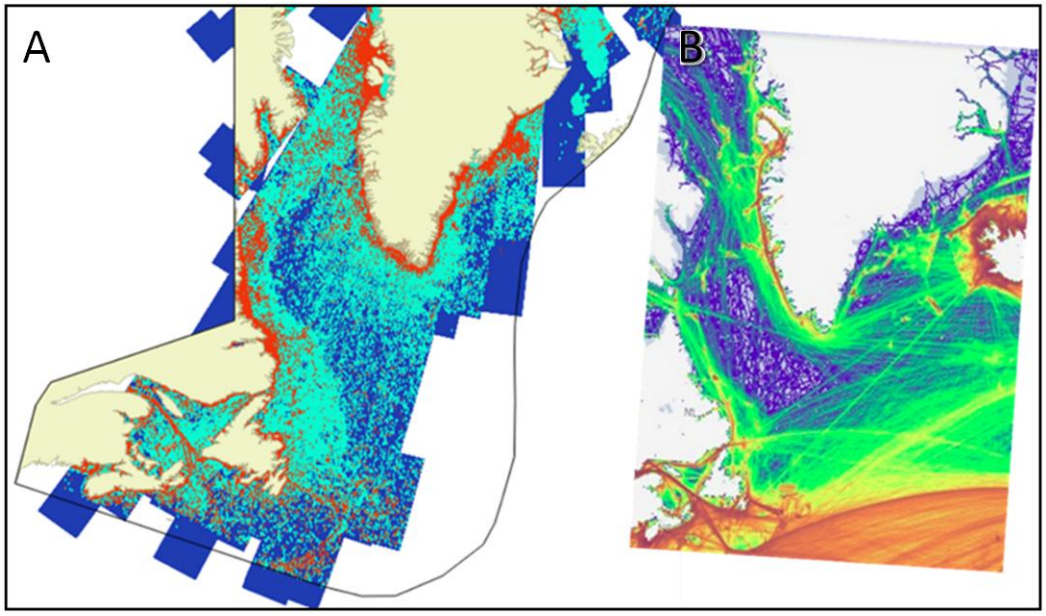

Figure 3: Plot of all targets detected in Sentinel-1 imagery acquired in IW mode during 2019. Colours indicate different detected targets (Red: targets "length-to-width-ratio" > 2.5, likely ships; turquoise: targets "length-to-width-ratio" < 2.5, likely icebergs) (A). Dark blue shows the coverage of satellite scenes (IW) overlapping the CMEMS area of interest for icebergs, i.e., the area inside the polygon which is outlined by the thin black line. Marine Traffic Density MAP (TDM) inferred from AIS ship tracking data (B) (source: marinetraffic.com).

Figure 3 shows that many areas and structures of dense ship traffic can also be identified as areas with large amounts of targets with high length-to-width-ratios ([target length] ^{2} / [target area] > 2.5) indicated by red in Figure 3A.

Although this simple relationship is not good enough to be used in a straightforward way as a ship-iceberg discriminator, it does explain some of the differences between ships and icebergs, which are also confirmed through manual quality checks carried out by the DMI Ice Service.

IV.2 The addition of RCMLN imagery

As part of the 2025 update, the DIDA processing chain was extended to include iceberg detections based on imagery from the RADARSAT Constellation Mission (RCM), beginning with the Low Noise (RCMLN) acquisition mode. This decision was supported by a comparative analysis (see tables 3a-b) using Sentinel-1 IW data as reference, due to its higher spatial resolution and reliability in iceberg detection. Tables 3a-b shows validation statistics for both RCMLN imagery and RCM ScanSARMedium-50 imagery vs. Sentinel-1 IW reference imagery. Interpretation of the tables is exemplified by the following:

If more icebergs- or an equal amount of icebergs are detected in a buffer zone when DIDA is applied on Sentinel-1 IW imagery than when it is applied on RCM imagery then Count_S1_IW ≥ count_RCM. In these cases the = and > signs tell whether the number of icebergs detected inside a buffer zone is the same in the RCM-test imagery or more targets were detected in the reference imagery (i.e., > means missing targets in the RCM-test imagery). Thus, for RCM Low noise, in 60 % of the cases (where there were no obvious false targets), DIDA yielded the same result as if it was applied on the reference imagery. In 40 % of these cases there were missing targets.

Table 3a. *Statistics on iceberg detections in RCM Low Noise imagery vs Sentinel-1 IW reference imagery: The terms count_S1_IW and count_RCM refer to the numbers of icebergs counted inside the buffer zones around each reference target.*

RCM imagery:	***Cases where:*** ***count_S1_IW = count_RCM*** ***OR count_S1_IW > count_RCM***		***Percentage of total iceberg detections where:*** ***count_S1_IW = count_RCM*** ***OR count_S1_IW > count_RCM***
RCM Low Noise (Raw backscatter)	*60% are = (equal to)*	*40% are > (larger than)*	*99 (i.e, 1% are obviously false targets)*
RCM Low Noise (Denoised)	*47% are = (equal to)*	*53% are > (larger than)*	*99 (i.e, 1% are obviously false targets)*

Table 3b. *Statistics on iceberg detections in RCM ScanSARMedium-50 vs Sentinel-1 IW reference imagery: The terms count_S1_IW and count_RCM refer to the numbers of icebergs counted inside the buffer zones around each reference target.*

RCM imagery:	***Cases where:*** ***count_S1_IW = count_RCM*** ***OR count_S1_IW > count_RCM***		***Percentage of total iceberg detections where:*** ***count_S1_IW = count_RCM*** ***OR count_S1_IW > count_RCM***
RCM ScanSARMedium-50 (Raw backscatter)	*19% are = (equal to)*	*81% are > (larger than)*	*78 (i.e, 22% are obviously false targets)*
RCM ScanSARMedium-50 (Denoised)	*1% are = (equal to)*	*99% are > (larger than)*	*80 (i.e, 20% are obviously false targets)*

A total of 4,988 reference iceberg targets from co-located Sentinel-1 IW imagery were identified and used to evaluate detection performance in corresponding RCM scenes. Within a 1 km buffer zone

around each reference target, iceberg concentrations were compared between the Sentinel-1 IW reference and the RCM test scenes.

Given that RCM imagery generally has lower spatial resolution, cases where fewer detections occur in RCM than in the reference are expected and typically reflect small or closely spaced icebergs not resolved in the RCM data. However, cases where more iceberg detections appear in RCM than in Sentinel-1 IW—despite the lower resolution—are considered to include obvious false alarms.

The analysis showed that RCM Low Noise imagery, particularly in its raw (non-denoised) form, performed substantially better than other RCM modes. In 60% of the buffer zones, DIDA returned the same number of detections in RCMLN as in Sentinel-1 IW, while 40% showed fewer targets, as expected. Only 1% of detections were classified as obvious false alarms—cases where more targets were detected in the RCM imagery than in the higher-resolution reference.

By contrast, ScanSARMedium-50 imagery showed much poorer performance, with up to 22% of detection cases falling into the false-alarm category and almost no instances of perfectly matching target counts with the reference.

These findings show that with no modifications to the present DIDA-settings RCMLN data provides the most balanced performance in terms of detection accuracy and minimal false positives, making it a natural starting point for integrating RCM into DIDA operations (as it could be incorporated with no adjustments to DIDA). However, on-going studies on noise correction and further adjustments to DIDA (e.g., a changed CFAR threshold regime) are expected to support the gradual inclusion of additional RCM acquisition modes in the coming years.

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

This section is dedicated to describe and track changes that may occur in the current operational system impacting product quality. This includes changes due to outages (data gaps), upstream data (e.g., addition or loss of a sensor, inclusion/removal of in-situ platforms), reprocessing, corrections, assimilated data, boundary conditions, among others.

Table 3: List of changes in the SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_007 production line.

Date	Change/Event description	System version	Other
30/11/2023	Introduction of new sea ice and ship filtering.	2.12	
30/11/2024	Expansion of geographical coverage for iceberg point positions datasets (shapefiles) to pan-Arctic coverage.	2.13	
30/11/2025	Expansion of geographical coverage for gridded datasets (netCDF files) to pan-Arctic coverage. This also involved change of grid-projection from Lambert Conformal Conic to polar stereographic. Three new datasets (2 x netCDF + 1 x Shapefile) based on RCM imagery were added.	2.14	

VI QUALITY CHANGES SINCE PREVIOUS VERSION

Since the previous release (November 2024), the iceberg dataset portfolio (as of November 2025) has been updated to include observations derived from RADARSAT Constellation Mission (RCM) imagery (three new datasets), in addition to the existing Sentinel-1 based products. The integration currently includes RCM Low Noise (RCMLN) imagery, which was selected for initial implementation based on its relatively low rate of “obviously false detections” in comparison with other RCM acquisition modes, when assessed against Sentinel-1 IW reference observations.

In parallel, the geographic coverage has been expanded from a focus on Greenland waters to a pan-Arctic domain – also for gridded products. This extension is enabled by updates to the DIDA (DMI Iceberg Detection Algorithm) processing chain and improved data access. These changes are expected to enhance the overall applicability of the product for operational and scientific users requiring broader spatial coverage.

VII REFERENCES

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